NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026
PYTHON FOR DATA ANALYTICS
PROJECT SUBMISSION

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TITLE	**: Food Delivery Sales & Customer Insights**
GITHUB LINK	**:** https://colab.research.google.com/drive/1HjPPLPoL5d8ZAiUwQ6Xomx2UcHtpK6BB?usp=sharing
DATE	**:**

1. ABSTRACT

Project Overview

This project analyzes a Food Delivery Sales dataset containing information about customer orders, restaurant names, food items, delivery time, ratings, and order values. The main objective is to understand customer behavior and identify sales trends in food delivery services.

The dataset was analyzed using Python libraries such as Pandas and Matplotlib in Google Colab. Data cleaning techniques were applied to handle missing values and improve data quality. Exploratory Data Analysis (EDA) was performed to study order patterns, customer ratings, popular food items, restaurant performance, and delivery times.

The analysis revealed that certain restaurants generated higher sales, some food items were ordered more frequently, and delivery time had a significant impact on customer

satisfaction. Visualizations such as bar charts and pie charts were used to present the findings clearly.

The project helps food delivery businesses make better decisions regarding menu planning, delivery management, and customer service. The final outcome provides valuable insights for improving customer satisfaction and increasing overall sales.

2. PROBLEM STATEMENT

Business Problem / Real-World Challenge

Food delivery companies receive thousands of orders every day. Understanding customer preferences, restaurant performance, and delivery efficiency is essential for business growth. Without proper analysis, companies may face issues such as delayed deliveries, low customer satisfaction, and reduced sales.

Data analysis helps identify popular food items, peak order periods, and customer behavior patterns. By analyzing the data, businesses can improve delivery operations, increase customer satisfaction, and maximize revenue.

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3. OBJECTIVES

Analyze food delivery sales data.

Identify top-performing restaurants.

Find the most ordered food items.

Study customer ratings and satisfaction.

Analyze delivery time performance.

Generate meaningful insights using data visualization.

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4. PROJECT SCOPE

This project focuses on analyzing food delivery sales data using Python in Google Colab. The project includes data loading, cleaning, analysis, visualization, and interpretation of results to support data-driven decision making.

5. TECHNOLOGIES USED

Development Environment

Google Colab

Python

Libraries Used

Pandas – Data loading and analysis

Matplotlib – Data visualization

NumPy – Numerical operations

Dataset Source

CSV File (Food Delivery Dataset)

6. APPLICATION FLOW

Step-by-Step Workflow

Load the food delivery dataset into Google Colab.

Import required Python libraries.

Clean and preprocess the dataset.

Analyze sales and customer-related information.

Create charts and visualizations.

Interpret results and generate insights.

Prepare the final report.

7. KEY PYTHON DATA ANALYTICS CONCEPTS USED

1. Data Handling

Reading and managing the dataset using Pandas.

2. Data Cleaning

Removing missing values and correcting data inconsistencies.

3. Exploratory Data Analysis (EDA)

Analyzing sales trends, customer ratings, and restaurant performance.

4. Visualization

Creating bar charts, pie charts, and graphs using Matplotlib.

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8. EXPECTED OUTPUT

Dataset preview

Cleaned dataset

Top restaurants chart

Most ordered food items chart

Customer rating analysis

Delivery time analysis

Sales performance visualization

(Insert Google Colab screenshots here.

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9. CHALLENGES FACED

Dataset preview

Cleaned dataset

Top restaurants chart

Most ordered food items chart

Customer rating analysis

Delivery time analysis

Sales performance visualization

10. LEARNING OUTCOMES

Technical Learning

Python programming

Pandas for data analysis

Data cleaning techniques

Data visualization using Matplotlib

Practical Experience

Working with real-world datasets

Generating business insights

Understanding customer behavior

Making data-driven decisions

11. CONCLUSION

The Food Delivery Sales & Customer Insights project successfully analyzed customer orders, restaurant performance, delivery efficiency, and sales trends. The insights obtained can help food delivery businesses improve customer satisfaction, optimize delivery operations, and increase revenue. This project demonstrates how Python-based data analytics can support effective business decision-making and provide valuable insights from large datasets.